

Sumir Srivastava

Full-stack software engineer with 3+ years of experience building enterprise-level data-driven applications using .NET, React, Python, and cloud platforms (Azure/AWS). Strong background in quantitative analytics systems, backend APIs, and machine learning projects including RAG and computer vision.

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SKILLS SUMMARY

- **Languages:** Python, C#, Typescript
- **Frameworks:** .NET 9, Node.js, React
- **Cloud:** Azure Cloud Services, AWS
- **Database:** Oracle DB, PostgreSQL, MySQL
- **Tools:** GIT, GitHub

EXPERIENCE

- **Shell India Markets Pvt Ltd.** Bengaluru
Associate Software Engineer - Full Stack Engineer (Full-time) August 2023 - Present
 - **Quant Portal (Full-Stack Market Risk Platform):** Owned backend and full-stack feature development for Quant Portal, a full-stack .NET enterprise Market Risk / Commodity Risk Management platform, supporting Quant Risk Analytics across multiple business lines within Shell. Tech Stack: .NET 9, JavaScript, AWS, and PostgreSQL.
 - **VaR What-if Scenario Engine:** Implemented a What-if scenario tool for Value at Risk (VaR) calculations, enabling users to simulate exposure and price changes and compute risk metrics effectively.
 - **VaR Breach Monitoring Dashboards:** Developed VaR breach monitoring dashboards across 15+ business entities to track and analyze risk metrics, facilitating proactive risk management and expanding coverage across Shell's global trading operations.
 - **Bulk Data Ingestion Pipelines & Backend APIs:** Developed bulk data pipelines using AWS supporting insert/update (upsert) operations for over 100,000 records across multiple modules, reducing manual data processing and improving data consistency and saving time per upload by 95%. Built and optimized REST APIs for large-scale risk analytics workflows.
 - **Audit-Enabled Correction Workflows:** Streamlined correction workflows by building entity-specific correction screens with full audit trails, ensuring regulatory compliance and reducing correction time by 30%.
 - **Stakeholder-Driven Feature Development:** Collaborated with quantitative risk analysts and traders globally to translate business requirements into scalable technical solutions for a business-critical internal platform.
- **Birlasoft Ltd.** Noida
Software Engineering Intern (Intern) May 2022 - August 2022
 - Created CI/CD pipelines on Azure DevOps to integrate various tools and services.
 - Implemented security measures, including software composition analysis (SCA), static application security testing (SAST) and dynamic application security testing (DAST).

PROJECTS

- **LLM-Assisted Lightweight RAG Pipeline (Python):** *GitHub Repo* - Built a local Retrieval-Augmented Generation (RAG) pipeline in Python to extract scheduling and monitoring data from PDFs using Large Language Models. Implemented lexical scoring, structural pattern boosts (dates/terms), and LLM-assisted query expansion for reliable document analysis. Tech: Python, LLMs, prompt engineering, embeddings & vector search.
- **AI-Based Real-Time Vehicle Speed Detection (YOLO & SSD, Python):** *GitHub Repo* - Developed a real-time vehicle speed detection system using deep learning models (YOLO and Single Shot MultiBox Detector - SSD). Processed video streams to detect vehicles, estimate speed, and generate tagged outputs with violation indicators. Tech: Python, YOLOv5, OpenCV
- **Personal Portfolio (React + TypeScript):** *GitHub Repo* - Designed and developed a responsive personal portfolio website using React and TypeScript to showcase projects, technical skills, and experience.
- **WayLane - Lane Detection using OpenCV (Python):** *GitHub Repo* - Built a lane detection system using Python and OpenCV that processes live and recorded video streams. Applied Canny Edge Detection and Hough Transform to identify road lane boundaries for small-scale autonomous navigation use cases. Tech: Python, OpenCV

EDUCATION

- **Amity University, Noida** Uttar Pradesh, India
Bachelor of Technology - Computer Science; July 2019 - June 2023
Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases

CERTIFICATIONS

- **Azure Cloud:** Cloud Fundamentals, Data Fundamentals and AI fundamentals - *view certificate*
- **Cisco CCNA:** Introduction to Networks - *view certificate*
- **AWS Cloud Practitioner:** Cloud Essentials - *view certificate*

PUBLICATIONS

- **Research Paper: AI based real time Vehicle speed detection using Deep Learning:**
Taylor and Francis Journal(Click to view) Authored a research paper on developing a system using deep learning techniques, specifically Single Shot MultiBox Detector (SSD), to accurately detect vehicle speeds in real-time.
- **Research Paper: Path Detection for Self-Driving Carts by using Canny Edge Detection Algorithm:**
IEEE(Click to view) Conducted research and developed a system using the Canny Edge Detection Algorithm to enhance path detection for self-driving carts.